

Python

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^{or} Reference: To automate boring stuff with python

Basics

• Expressions

Example

> 2+2
4

Operator

**	Exponent	$2 ** 3 \Rightarrow 8$
%	Modulus	$22 \% 8 \Rightarrow 6$
//	floor division	$22 // 8 \Rightarrow 2$
/	division	$22 / 8 \Rightarrow 2.75$
*	Multiplication	$3 * 5 \Rightarrow 15$
-	Subtraction	$5 - 2 \Rightarrow 3$
+	addition	$5 + 2 \Rightarrow 7$

String concatenation

• 'Alice' + 'bob' $\Rightarrow$ 'Alicebob'

Replication

• 'Alice' * 3
 $\Rightarrow$ 'AliceAliceAlice'

• Variables

a container where you can store data

Ex: x = 144

Rules

- Hyphen are not allowed
- Space X
- Can't begin with nr.
- Special character X

Data Types

- Integer : whole number . Ex : 69
- Float : decimal point number : Ex : 69.69
- String : represents text

Flow control ← deciding the orders in which how program runs as decided.

<u>Comparators</u>			
$=$	equal	$2 = 2$	True
$\neq$	not equal	$2 \neq 2$	False
$>$	greater than	$2 > 1$	True
$<$	less than	$1 < 1$	False
$\geq$	greater equal	$2 \geq 2$	True
$\leq$	less equal	$3 \leq 1$	False

Boolean

- and True if both condn $(5 > 3) \text{ and } (10 > 5)$ True
- or True if one condn $(5 > 3) \text{ or } (1 > 3)$ True
- not rev. condition $\text{not}(5 > 3)$ False

if statement
runs a block of code if conditions are True
Ex:

```
age = 19
if age >= 18:
    print('you are adult')
```

elif [if else] [if you wanna check more than one condition]
Ex:

```
elif age < 18:
    print('you are a minor')
```

else [if no conditions are true]
Ex:

```
else:
    print('invalid')
```

while loop [Repeats code of block as long as cond is True].
Ex:

```
while spam < 5:
    print('hello world')
    spam += 1
```

break [condition true then break from loop]
Ex:

```
while True:
    name = input('name:')
    if name == 'your name':
        break
    print('Thank you')
```

continue [ends the current iteration of that a true condition in loop]
Ex:

```
i = 0
while i < 10:
    if i % 2 == 0:
        i += 1
        continue
    i += 1
    print(i)
```

print: 1
3
5
7
9

while not: [Reverse while condition]

For loop

- lets you iterate (go through) a sequence like:

```
for i in range(5):
    print('Timmy' + str(i+1) + '')
```

enumerate: used to loop through an iterable (list, tuple, string)

Modules

is a python file (.py) that contains code (functions, variables, classes) you can reuse.

Example: import: command to import modules in code
modules ex: random, math etc.

Functions

- is a block of code that performs a specific task and be reused multiples times.

Syntax: `def functionname(parameters):`

Call stack

- like a stack of plates.
- when function is called, python pushes onto stack.
- when function finishes, python pops it off the stack.
- always run the topmost function.

Ex:

```
def fna():
    print("Inside A")
    fnc()
    print("Back to A")
    fnc()
def fnc():
    print("inside B")
    fnc()
def fnc():
    print("inside C")
fna()
```

• Local & Global Scope

• Local Scope

Variable inside a function

Exist only while the function runs

Ex:

```
def spam():
    egg = 69
eggs = 9
```



both are diff.

• Global scope

: Variable declared outside any function

global egg = 99

bacon = 33

Shadowing

happens when a local variable has the same name as a global variable

here the local scope shadows the global one inside that func.

Exception handling

exception is an error that occurs during program execution. Not handled python stops program.

• try / except

try :

`n = int(input("number"))`

`print(n)`

except ValueError :

`print("invalid")`

• Practice Problem

⇒ Zig Zag

⇒ Collatz Sequence

x

Collatz problem

→ `func : collatz → [parameter number]`

→ if no. is even → Ans `number // 2`

→ if no. odd → Ans `3 * number + 1`

• keep calling till it returns value 1.

list

- list is a set of data which is mutable.

Ex = $i[4] = ['cat', 'dog', 'mouse', 'hoo']$
 0 1 2 3

changing value in list

$i[1] = 'hello'$

∴ in index hello is present & dog removed

→ can perform concatenation $list1 + list2$

→ can perform multiplication $list1 * 3$

- using `del` = to perform delete an element in the list

- Using loop

$list = [1, 2, 3, 4, 5]$, $(n-1)$

→ for i in range $[0, n)$:

$print(i)$ → print all value from 0 to $(n-1)$.

→ if i in list:

$print('Present')$ → using in method

→ Using enumerate

$list = ['A', 'B', 'C', 'D', 'E']$

for index, item in enumerate(list);

$print('Index' + str(index) + 'is' + item)$

→ Using random module

~~list~~ import random

~~list~~ $list = ['A', 'B', 'C', 'D', 'E']$

$random.choice(list)$

- `random.shuffle(list)` # shuffles the list

Augment assignment operator

- `spam += 4`
- `spam -= 4`
- `spam *= 4`
- `spam /= 4`
- `spam %= 4`

Methods

- `list = ['A', 'B', 'C', 'D', 'E']`
- `list.index('B')` # it will give you one 1.
- `list.append('F')` # gonna add element at end of list
- `list.insert(1, 'A1')` # gonna insert element at desired index
- `spam.remove('A1')` # remove the element
- `spam.sort()` # sorts list
- `spam.sort(reverse = True)` # descending type

Practice problems

- Magic 8 ball / fortune teller # To minimize program length use list

Sequence data types

example

name = 'Zophie'

name[0] = 'Z'

→ lists are mutable

→ String is immutable

Tuples

tuples are immutable

list = ['A', 'B', 'C', 'D', 'E']

list[0] = 'Z'

error ↑

→ Same as list only here it is immutable.

Understanding id function

id → returns unique identifier (memory address) of an object.

Practice Question

1. Comma Code ✓

spam = ['apples', 'bananas', 'tofu', 'cats']

2. Conclip Streaks ✓

Dictionary

Ex: `my_cat = {'size': 'fat', 'color': 'gray', 'disposition': 'loud'}`
 ↓ ↓
 key value

Methods

1. `keys()` = # print values of key in dictionary
 Ex: `cat.keys()` [# in loop]
2. `items()` = # print value key pair value.
 Ex: `for i in cat.items():`
3. `keys()` = # gives us the key in dictionary -
4. `get()` = return ^{given} key's value and if not present return other value.
 Ex: `cat.get('size', 0)`
5. `setdefault()` = if given key is present then returns it value or not then gets added in dictionary with new value.
 Ex: `cat.setdefault('color', 'black')`

Project

- simple tic tac toe program game ✓
- chess dictionary validator ☉ ✓

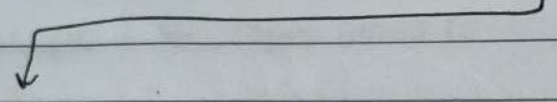
• `strip()` → removes white space

• `ord('A') = 65`

• `chr(65) = A`

Using Pyperclip

- Project
- multi-clipboard Automatic message
- Pig latin



Using sys module

`sys.argv`

→ Terminal is a list the word you typed in terminal

Three Brainstorm

1v1

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Tic tac toe

Board = { }

↓
pair (strutem)

↓
while True

while choose no. 1 to 9

change

1 (end)

TL	TM	TR
1	2	3
4	5	6
7	8	9
LF	LM	LR



✓ TL = TM = TR

✓ TL = ML = LF

✓ LF = L = LR

✓ TL = TM = TR

1v1 human

def win rule (

✓ TR, MR, LR

✓ TR, M, MR

✓ TM, M, ML

✓ LF, M, TR

✓ TL, M, LR

while True (

turn X =

Multi-clipboard

import pyperclip, sys

↓
text = { '...', '...', '...' }

↓
if len(argv) < 2:

print error

sys.exit

↓
keyphrase = sys.argv[1]

pyperclip.copy(text[keyphrase])

print('successfully copied')

else: Not successful